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Prediction of significant wave height using machine learning and its application to extreme wave analysis

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Waves of large size can damage offshore infrastructures and affect marine facilities. In coastal engineering studies, it is essential to have the probability estimates of the most extreme wave height expected during the lifetime of the structure. This study predicts significant wave height using the machine learning (ML) technique with generalized extreme value (GEV) theory and its application to extreme wave analysis. The wind speed, wind direction, sea temperature, and swell height data consisting of wave characteristics for 60 years has been obtained from the European Centre for Medium-Range Weather Forecasts (ECMWF). While analyzing extreme waves, the block maxima approach in GEV was used to incorporate the seasonal variations present in the data. The estimated scale parameter, shape parameter, and location parameter of GEV are used in the ML model to predict the significant wave heights along with their return periods. The ML algorithms such as linear regression (LR), artificial neural networks (ANN), and support vector machines (SVM) are evaluated in terms of R^2 performance. The model comparison results suggested that the SVM model outperforms the LR and ANN models with an accuracy of 99.80%. Finally, the GEV analysis gives the extreme wave height results of 2.348, 3.470, and 4.713 m with a return period of 5, 20, and 100 yrs, respectively. Hence, the model developed is capable of predicting both significant wave height and extreme waves for the design of coastal structures.

Keywords. Significant wave height; generalized extreme value; artificial neural network; support vector machine; machine learning.

1. Introduction

Water bodies cover 71% of the earth's surface area, with the ocean contributing 96.5% of water. The ocean is a great storehouse of energy; the movement of waves across the surface of the water body is affected by various parameters such as wind speed, wind direction, wave speed and gravitational forces. The design of offshore structures requires an accurate estimation of these extreme wave parameters. In engineering applications, extreme wave height and wind speed are considered important factors in the design of coastal and offshore structures, ship routing and the operation of facilities. Therefore, the prediction of significant wave height can be used in many areas including finding ideal spots for tidal energy generation, route planning for different ships, forecasting

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currents, finding fishing holes, construction of offshore structures such as bridges, oil rigs and many other examples as well.

Extreme wave conditions influence the design of offshore structures and their expected design life. The knowledge of extreme waves with a return period of 50 or 100 yrs is important for designing the offshore structure. Therefore, extreme value analysis becomes essential for the design of the offshore structure. Generally, the extreme value analysis is carried out by generalized extreme value (GEV) theory and generalized Pareto distribution (GPD). Extreme value analysis has a number of applications in environmental forecasting. Palutikof *et al.* (1999) conducted an extreme value analysis on wind speed. Similar work was conducted by Lin (2003) on wind gusts, and Emanuel and Jagger (2010) on hurricane data, respectively. As mentioned earlier, for the construction of an offshore structure at any geographical location, it is very important to study the extreme waves that the structure needs to deal with during its design life. Therefore, accurate prediction of wave heights is required to manage or safeguard the coastal structures.

Conventional numerical modelling tools such as simulating waves nearshore (SWAN) (Booij *et al.* 1996), MIKE 21 (Warren and Bach 1992) and DELFT-3D (Hydraulics 1999) are used to predict the significant wave height. Recently, Afzal and Kumar (2021) used a third-generation numerical model to predict wave parameters in coastal regions, but the numerical model used is more complex and time-consuming. In recent times, machine learning (ML) tools such as linear regression (LR), artificial neural network (ANN), support vector machine (SVM) etc. have been adopted to predict wave parameters, which gives better prediction result with less computation time with respect to the conventional model. The ML models are mainly data-driven but the ability of ML models to learn from experience means they can also learn physics (Dutta *et al.* 2020; Kumar *et al.* 2020). This indicates that if enough samples of a physical system's behaviour are presented, ML models may learn it and make predictions based on it. A class of these models called ANN is inspired by the way the human brain processes information and learns from experience.

In recent years, wave parameters are predicted frequently using the ANN technique by several researchers (Asma *et al.* 2012; Vimala *et al.* 2014; Tsai *et al.* 2017). These studies showed that ANN is relatively better than traditional numerical-based models. As the number of hidden layers and neurons directly influences the performance of neural networks, finding the optimum number of hidden layers and neurons can be achieved through experience and trial and error. Mahjoobi *et al.* (2008) used the fuzzy inference system (FIS) and adaptive-network-based fuzzy inference system (ANFIS) models to predict wave parameters, which concluded a marginal accuracy over ANN. Dehghan *et al.* (2009) proposed a numerical simulation of the Helmholtz equation by using modified Hopfield neural networks. James *et al.* (2018) used SVM for the classification of the period of peak wave. Mahjoobi and Mosabbeb (2009) applied the SVM model to predict the significant wave height for the Lake Michigan data. Fan *et al.* (2020) predicted the significant wave height using a novel long short-term memory (LSTM) network. In the ML technique, the radial basis function is used to improve or tune the prediction accuracy. Kalra *et al.* (2005) show the estimation of significant wave heights based on the concept of RBF and ANN, while Nourani and Babakhani (2012) discussed the combination of ANN with RBF interpolation for modelling earthfill dam seepage.

In recent times, conventional models are coupled with ML models to predict significant wave height. Londhe *et al.* (2016) developed a numerical model coupled with an ANN to improve the wave forecast of a specific location. They used the dataset of numerical wave forecasts made by INCOIS and further coupled it with the ANN model at four stations along the Indian coastline. They observed that the coupled numerical with ANN model gives better and more accurate prediction as compared to numerical models. Further, Deshmukh *et al.* (2016) proposed an approach to reduce the errors from numerical model results as compared to the observational dataset. They coupled the numerical model with a wavelet ANN to improve numerical ocean wave predictions of significant wave height and peak wave period. Recently, Zhang *et al.* (2021) used SWAN model output to develop numerical long short-term memory (N-LSTM) to predict ocean wave height.

The conventional methods used to study extreme wave conditions and significant wave height prediction were complex and time-consuming. The introduction of ML algorithms in the field of civil and coastal engineering made it possible to overcome these problems. Therefore, the objective of the present study is to analyze extreme wave

conditions and predict the significant wave height by combining ML and concepts of GEV theory. To the authors' knowledge, there does not exist a study that dealt with both of these issues at the same time. This paper illustrates significant wave height prediction using the SVM technique, which is a relatively new approach as compared to ANN.

2. Numerical model

2.1 *Input data*

The study area of the present study is Mehamn harbour located in Norway (the map location is marked in figure 1). The latitude and longitude of Mehamn harbour are 71.19° and 28.03°, respectively. The wave data belonging to Mehamn harbour consists of the data for a period of 60 yrs starting from September 1st, 1957 (6 AM) to May 31st, 2017 (6 PM) with an interval of 3 hrs each. Generally, the wind speed in the study region varies from 5 to 14 m/s in the south-southwest direction. In Mehamn Harbour, the significant wave height varies from 2.4 to 3.2 m for the peak wave period of 9–12 s. However, the swell height varies from 1.5 to 1.9 m. Further, the wind, wind sea and swell parameters datasets are obtained from the WAM model developed by the WAM Development and Implementation Group, led by Klaus Hasselmann (Group 1988). The WAM model-generated data were used as input to ML models. The global stand-alone wave model (WAM) of the European Centre for Medium Range Weather Forecasting (ECMWF) reanalysis dataset is used in the present study. The WAM model shows excellent performance on global and regional scales with field observation (Luo and Flather 1997). The WAM model runs on a spherical latitude-longitude grid and can be used in any ocean region. Several studies used the WAM model data and validated it with *in-situ* observation at different locations (Janssen *et al.* 1989; Bauer *et al.* 1992; Musić and Nicković 2008). To the best of the authors' knowledge, there is no field data available

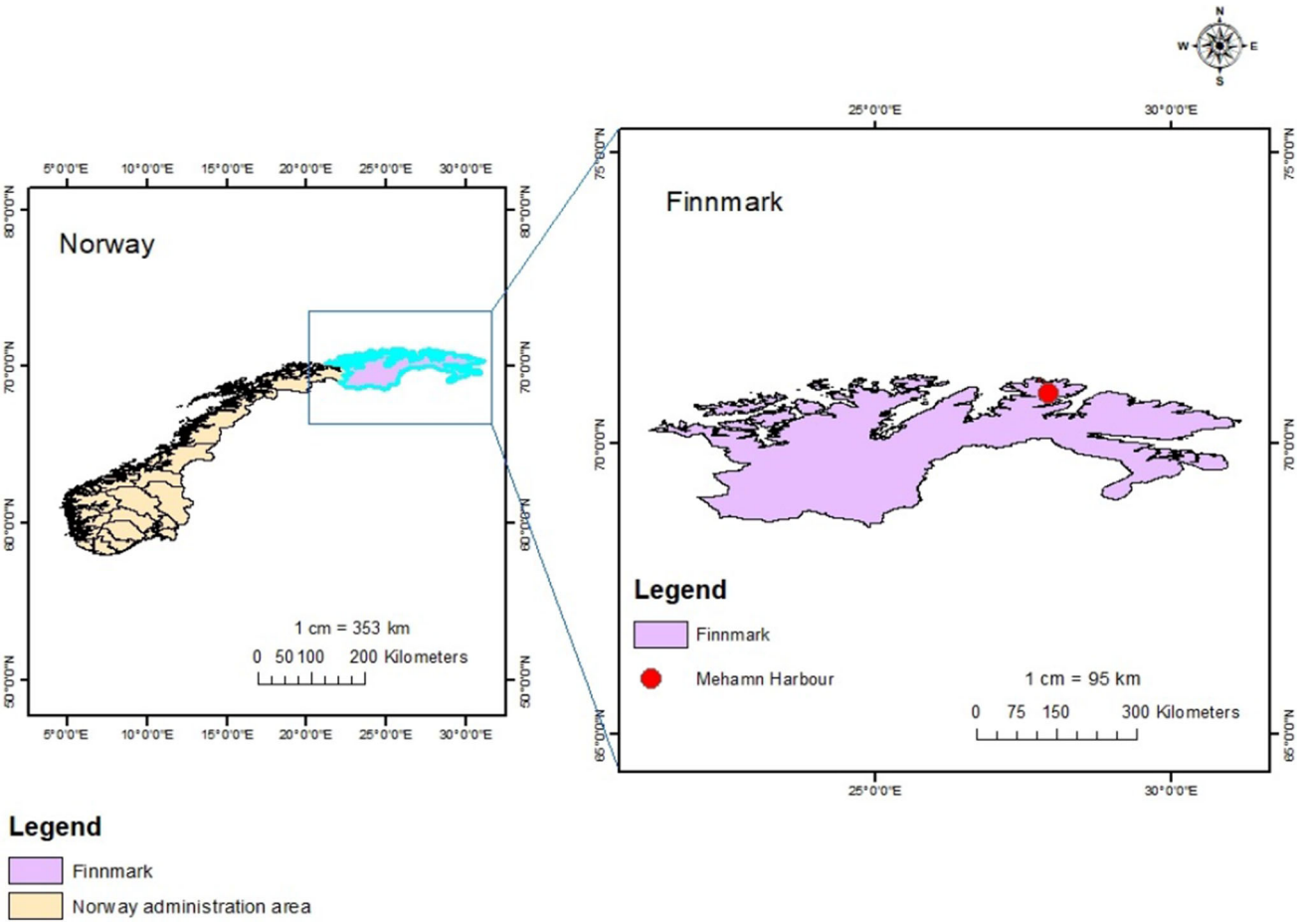


Figure 1. Location map of Mehamn harbour.

either in literature or with any agency that collects wave buoy data of Mehamn harbour. Secondly, the data that has been used in the present model, i.e., WAM data from ECMWF is model-generated data. This model had been validated extensively and the generated data has acceptance by the scientific community worldwide, i.e., considered equivalent to measured/field data itself. Thirdly, the huge WAM data is divided into two parts while preparing the machine learning model. The first set (50% of the data, i.e., 1957–1986 data) had been used to prepare/train the machine learning model. After that, the second half of the data (i.e., 1987–2017) had been used as validation data against the predictions done by the developed machine learning model for the same period, i.e., 1987–2017. This is a standard practice in machine learning, i.e., testing which is considered the same as the validation of the numerical model.

ERA-40 and ERA-Interim are the reanalysis dataset of the global atmosphere and surface conditions for 45 yrs, over the period from September 1957 through August 2002 by ECMWF. ERA-40 follows the ERA-15 and ERA-40 reanalysis dataset. ERA-Interim uses a 12-h, 4-dimensional variance analysis (4D Var) focused on the ECMWF with an efficient estimate of differences in satellite radiance results (Var BC). ERA-40 (Uppala *et al.* 2005) was a 2nd generation reanalysis dataset, which was produced by ECMWF. The 3D-variational analysis was used as an assimilation method in the ERA-40 dataset. ERA-40 dataset uses the spectral grid with triangular truncation of waves (corresponds to about 125 km) and a hybrid vertical coordinate scheme of 60 levels. The ERA-40 data used in this analysis were collected from the ECMWF server on a 2.5° × 2.5° resolution of the fixed grid (corresponding to around 250 km). The 2D output spectra include 25 wave frequencies and 12-direction bins.

ERA-Interim is a 3rd generation reanalysis data of ECMWF, which follows the ERA-15 and ERA-40 datasets. ERA-Interim uses a 12-h, 4-dimensional variance analysis (4D Var) with an accurate calculation of the satellite radiance effect variations (Var BC). It improves the quality of satellite observations (Dee *et al.* 2011). The spatial resolution of the dataset is 0.75° × 0.75°. The 2D output spectra include 30 wave frequencies and 24 direction bins. ERA-40 and ERA-Interim are reanalysis datasets, which do not require any cleansing or feature engineering. Therefore, ERA-40 and ERA-Interim reanalysis datasets of ECMWF are used in the WAM model to predict significant wave height.

In addition to atmospheric variables, the ERA-40 and ERA-Interim often provide wave parameters produced by a two-way atmosphere-wave model system in which wind fields and other atmospheric parameters that affect wave growth are transmitted to the wave model. WAM wave model is used to combine these control frameworks by incorporating the so-called wave energy balance equation, the wave model WAM emits 2D wave energy spectra at each grid point, and multiple combined wave parameters can be derived from the spectra. The wind speed and swell parameters can be obtained in the wave model WAM by combining the high and low-frequency portions of the wave spectra, respectively.

2.2 *Linear regression*

Linear regression is a linear model in which the input variables and the single output variable are assumed to have a linear relationship (Montgomery *et al.* 2021). It is based on a statistical approach, which is easier to implement, and interpret, and performs exceptionally well for linearly separable data. However, it is vulnerable to noise and overfitting. A linear regression (LR) model evaluates the best-fitting linear curve, for training data points, known as a hypothesis. A general hypothesis function (H) is given by equation (1).

$$H(x) = a_1 + a_2x_1 + a_3x_2 + a_4x_3 + ... + a_nx_{n-1}, \quad (1)$$

where $a_1, a_2, a_3, ..., a_n$, are constants of the best-fit curve. An example of a hypothesis function for a single input feature is shown in figure 2.

The LR technique may not be able to capture the nonlinear behaviour of the wave parameter. However, the ANN model is flexible enough to approximate non-linear relationships between variables without requiring *a priori* assumptions about their form of relationships. The LR model is discussed more in detail in the appendix section of the present study.

2.3 *Artificial neural networks (ANN)*

The artificial neural network (ANN) model uses the same approach as our brain does (Asma *et al.*

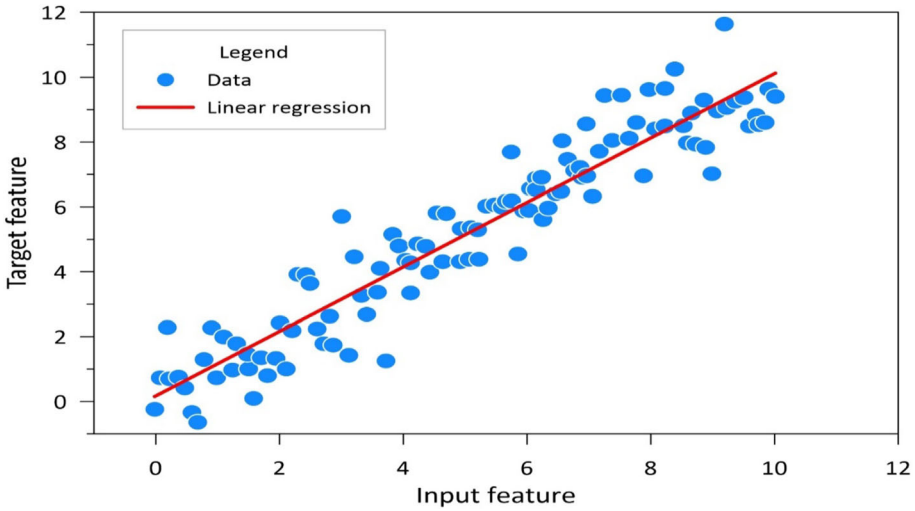


Figure 2. Hypothesis function for a single input feature.

2012). It is the advanced version of the LR technique that interprets the complex and non-linear relationships of input as well as output. The ANN learning methods are quite robust to noise in the training data. The training examples may contain errors, which do not affect the final output. ANNs can bear long training times depending on factors such as the number of weights in the network, the number of training examples considered, and the settings of various learning algorithm parameters. It is used where the fast evaluation of the learned target function is required. Despite their versatility, neural networks are prone to overfitting susceptible to local minima, which means they may fail to accomplish loss function minimization. To address this difficulty, researchers turned to the SVM technique. The ANN model is discussed more in detail in the appendix section of the present study.

2.4 *Support vector machines (SVM)*

Support vector machine (SVM) algorithm is a non-linear generalization of the Generalized Portrait algorithm developed in Russia in the sixties (Vapnik 1963; Vapnik and Chervonenkis 1965). SVMs are structured learning models that analyze data used for regression and classification analysis, with associated learning algorithms. SVM is based on non-linear statistical theory to transform input space into a higher dimensional feature space to separate the data patterns (Baylar *et al.* 2009). The goal of the SVM is to find an optimal hyperplane, which could differentiate the data in different classes by the maximum gap (Smola and Schölkopf 2004). The unique feature of SVM is that it uses kernel function to observe the non-linear behaviour of the parameter. However, LR and ANN model tries to fit a line to data by minimizing a cost function. The SVM does not perform very well when the data set has more noise, i.e., target classes are overlapping. In general, The SVM model is believed to be computationally inexpensive and efficient (Maier and Dandy 2004). The SVM model is discussed more in detail in the appendix section of the present study.

2.5 *Generalized extreme value theory (GEV)*

The generalized extreme value theory (GEV) distribution is a type of probabilistic approach used for modelling the smallest or largest set of independent random values or observations that are identically distributed (El Adlouni *et al.* 2007; Juma *et al.* 2021). In simpler terms to study and model the extreme ends of the graph of a frequency distribution curve, where though the frequency of the event to happen is significantly low but still poses a great deal of challenge to study and analyze the effect.

The GEV combines three simpler distributions in one form and allows a continuous range of different possible forms, including all three simple distributions. The author can use any of these distributions for modelling a particular data set of 'block maxima', which refers to the largest of entities in each batch or minima among the smaller ones. The GEV distribution also allows 'to let the

data decide' its appropriate distribution. Three simpler distributions of GEV distributions are referred to as Type I or Gumbel, Type II or Frechet and Type III or Weibull. Distributions whose 'tails' are finite, such as beta lead to Type III or Weibull distribution. The cumulative distribution function for GEV is given in equation (2).

$$F(x;\mu,\sigma,\varepsilon) = \exp\left\{-\left[1+\varepsilon\left(\frac{x-\mu}{\sigma}\right)\right]^{\frac{-1}{\varepsilon}}\right\} \text{ if } \varepsilon \neq 0,$$
$$F(x;\mu,\sigma,\varepsilon) = \exp\left\{-\exp-\left(\frac{x-\mu}{\sigma}\right)\right\} \text{ if } \varepsilon = 0. \quad (2)$$

The parameters, ε, μ, and σ, represent the shape, location, and scale of the distribution function, respectively.

The shape parameter ε, is a very important parameter in the analysis of extreme values as it controls the heaviness of the tail of the distribution, which in turn directly affects the extreme values. The shape parameter covers three different types of tail behaviour:

(1) For $\varepsilon > 0$: Frechet distribution,
(2) For $\varepsilon = 0$: Gumbell distribution,
(3) For $\varepsilon < 0$: Weibull distribution.

3. Methodology

Figure 3 shows a schematic flowchart depicting the relation between different components of the ML model developed in the study. The wind, wind direction, and swell parameters reanalysis data is obtained from European Centre for Medium-Range Weather Forecasts (ECMWF). The pre-processed reanalysis data are used due to the lack of observational data at the location. However, if such data is available, the observational data can also be used instead of the ECMWF for the development of this type of ML-based prediction model.

The implementation methodology of extreme wave analysis comprises three steps. The first step consists of ML model development to predict significant wave height using the training dataset (data from 1957–1987 in the present case). The second step is the prediction of significant wave height from 1988–2017 using the developed ML model and comparing it with the original data set obtained from ECMWF for the years 1987–2017. The third step involves the analysis of extreme wave conditions using the GEV model with ML-predicted significant wave height datasets from 1957–2017. First of all, the input dataset of wind, wind sea and swell parameters are obtained from ECMWF for Mehamn harbour, Norway. The wind speed (U) and wind directions (θ) are the wind parameters used in the present study. The wave height, peak wave period, and direction of peak wave of wind-generated wave are denoted by H_w, T_p, and θ_p, respectively. The swell height (H_{sw}) and swell direction (θ_s) are used as swell parameters in the present study.

Further, the dataset mentioned above is divided into training and testing sub-datasets in a 1:1 ratio. This means that 50 percent of the data set (i.e., 1957–1987) is used to train and develop the ML model, and the remaining 50 percent data set (i.e., 1988–2017) will be used for validation and testing of the model. In addition, the developed or trained ML model is used to predict the total significant wave height and their predictions are validated with the testing dataset (1988–2017). Thereafter, the predicted data set (using the ML model) from 1957 to 2017 is used as input to the GEV model. Finally, the GEV analysis gives the extreme wave height results with the corresponding return period.

3.1 *Prediction of significant wave height*

The data consists of wind, wind sea and swell parameters. This dataset is split into training and testing datasets in a 1:1 ratio. The training dataset is used to develop an SVM model using the kernel as a 'radial basis function'. The trained SVM model is then tested to predict the significant wave height of the testing dataset. The significant wave height is predicted using wind, wind sea and swell parameters as independent variables and significant wave height being the dependent one.

3.2 *K-fold cross-validation*

Cross-validation is a resampling process that is used for the assessment of ML algorithms on a limited data sample. The K-fold cross-validation technique is used here in order to deal with overfitting. The procedure has one parameter named k referring to the number of groups in which a specified data sample is divided. In K-fold cross-validation, data is analyzed K times, where for each K analysis, K–1 folds are used for training

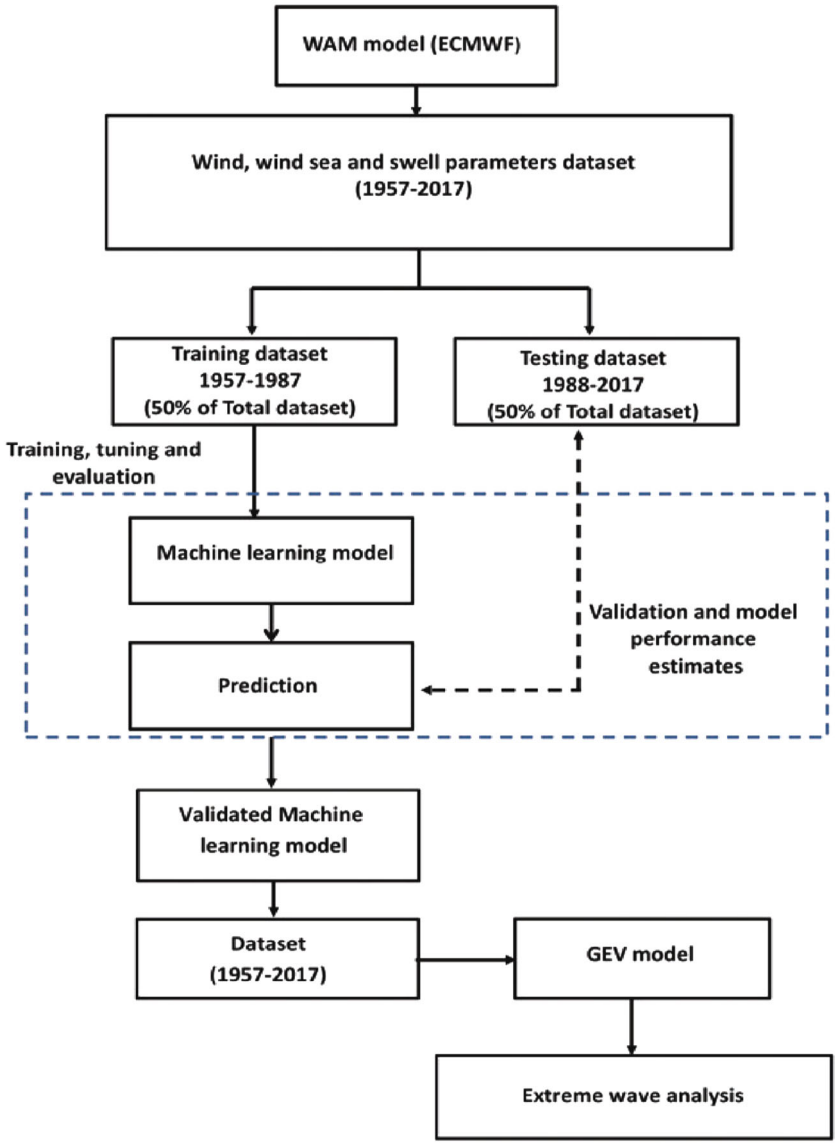


Figure 3. A flowchart illustrating relation between observation data, WAM data and machine learning model.

and the remaining fold is used as a testing dataset. The major advantage of performing K-fold cross-validation is that each observation is eventually used for both training and testing. In applied ML, cross-validation is primarily used to estimate an ML model's ability to learn from unseen data (Brownlee 2018). The dataset was divided into five folds. The model was tested on one of the folds using the remaining folds as training data, as shown in figure 4.

3.3 *Extreme wave analysis*

The frequency distribution (as shown in figure 5) shows the frequency of a particular significant wave height and thus the probability of its occurrence. Figure 5 depicts that most of the values of significant wave height lie between 0–2 m.

The prime objective of this study is to analyze the extreme wave conditions, which cannot be analyzed from the aforementioned plot. As a result, we need to model the tail of the density distribution (as shown in figure 6) in order to analyze the occurrence of extreme wave height probability using the generalized extreme value theory.

In extreme value theory (EVT), the block maxima (BM) technique divides the observation time into non-overlapping blocks of equal size and focuses on the largest observation in each period. The BM approach models the maximum

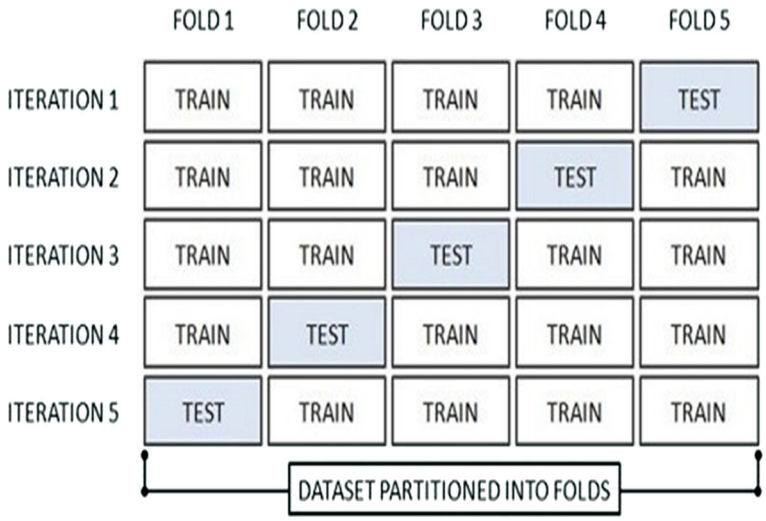


Figure 4. Splitting of data in train and test folds.

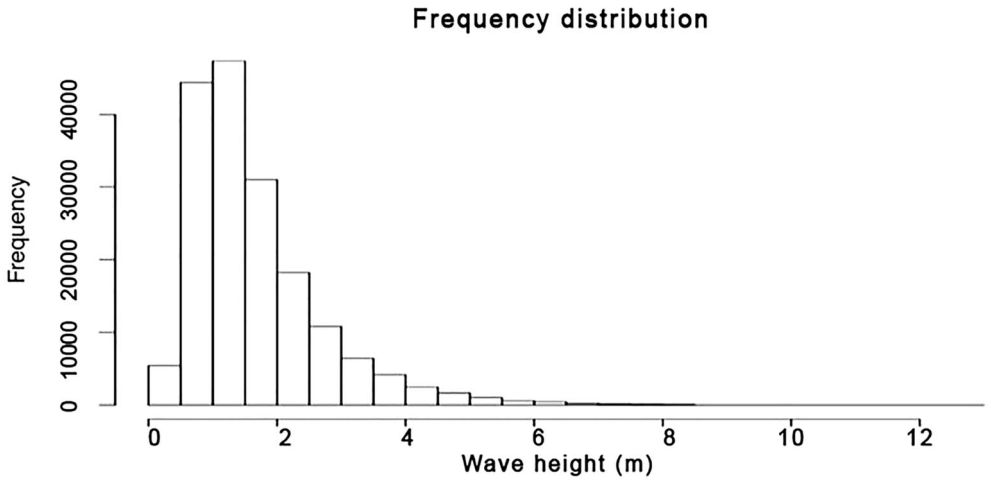


Figure 5. Frequency distribution of significant wave height data.

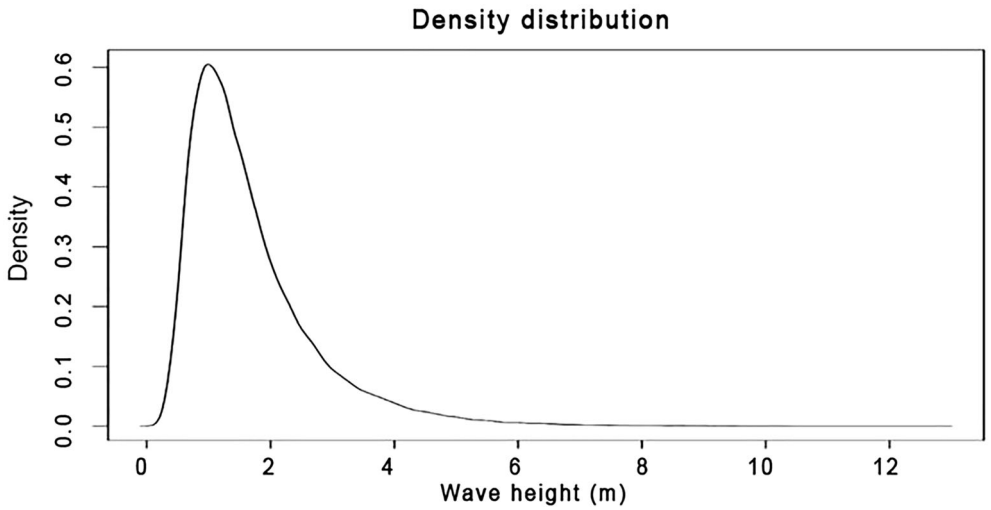


Figure 6. Density distribution of significant wave height data.

values of significant wave height from each time block. Since the dataset contains significant wave height information over a period of 60 yrs, figure 7 shows that the entire dataset was divided into 60 blocks and maxima from each of the blocks were segregated and used for model fitting. The block size was taken to be one year in order to incorporate any kind of variation due to seasonality.

The model fitting as depicted in figure 8 is processed and these 60 maxima points yielded the values of the shape, scale and location parameters, which are: $\varepsilon = -0.054$, $\mu = 7.432$, $\sigma = 1.124$.

These values of parameters were estimated based on maximum likelihood estimation since the observations for significant wave height were known. The value of the shape parameter corresponds to the Weibull distribution. Based on these values of GEV parameters, equation (3) is used to calculate the return level (i.e., extreme wave height (Z_T)) with a return period of 'T' years.

$$Z_T = \mu - \frac{\sigma}{\varepsilon}\left[1 - \left\{-\log\left(1 - \frac{1}{T}\right)\right\}^{-\varepsilon}\right] \text{ if } \varepsilon \neq 0,$$
$$Z_T = \mu - \sigma \log\left\{\log\left(1 - \frac{1}{T}\right)\right\} \text{ if } \varepsilon = 0. \quad (3)$$

The return period of a particular event is the inverse of the probability that the event will be exceeded in any given year, (i.e., $p = 1/T$). Substituting x with 'Z_T' and y with '1–p' in equation (2) provides the equation (3) as shown in equations (4 and 5).

$$y = \exp\left\{-\left[1 + \varepsilon\left(\frac{x-\mu}{\sigma}\right)\right]^{\frac{-1}{\varepsilon}}\right\} \text{ if } \varepsilon \neq 0,$$
$$1 - p = \exp\left\{-\left[1 + \varepsilon\left(\frac{x-\mu}{\sigma}\right)\right]^{\frac{-1}{\varepsilon}}\right\}$$
$$-1 + [-\log(1-p)]^{-\varepsilon} = \varepsilon\left(\frac{x-\mu}{\sigma}\right), \quad (4)$$
$$x = \mu - \frac{\sigma}{\varepsilon}\{1 - [-\log(1-p)]^{-\varepsilon}\}.$$

$$y = \exp\left\{-\exp -\left(\frac{x-\mu}{\sigma}\right)\right\} \text{ if } \varepsilon = 0,$$
$$1 - p = \exp\left\{-\exp -\left(\frac{x-\mu}{\sigma}\right)\right\}$$
$$\log(|\log(1-p)|) = -\frac{x-\mu}{\sigma}, \quad (5)$$
$$x = \mu - \sigma \log\{-\log(1-p)\}.$$

4. Results

The input data of Mehamn harbour contains wind, wind sea and swell parameters. These parameters have been used to predict the significant wave height using ML. Since ML includes a number of techniques and approaches, choosing an appropriate ML technique is an important task. Hence, a comparative study was done on Mehamn harbour data wherein the data was analyzed using different ML methods such as LR, ANN and SVM. The study also included the comparison of the performance of SVM by varying kernels.

Table 1 shows the comparison of RMSE and R^2 values for different ML models used in the present study. The LR and ANN models predict wave height with an accuracy of 98.90% and 99.40%,

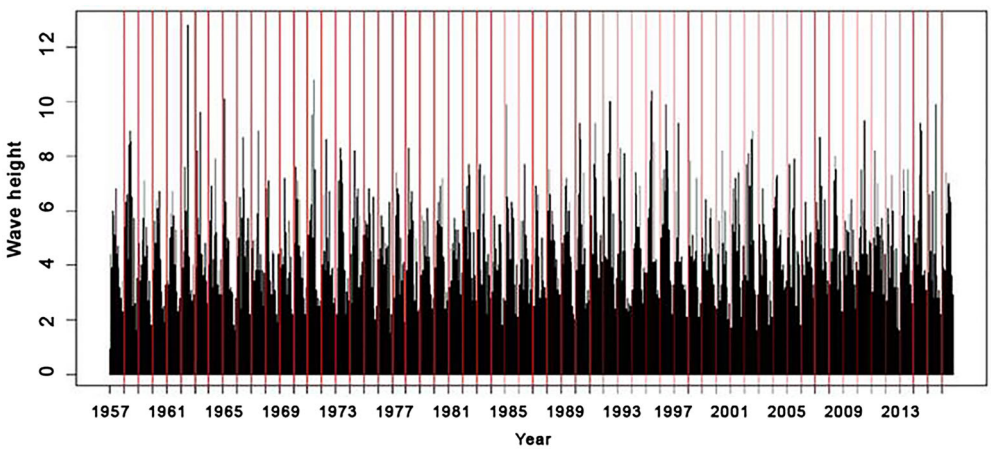


Figure 7. Data split into blocks of one year each.

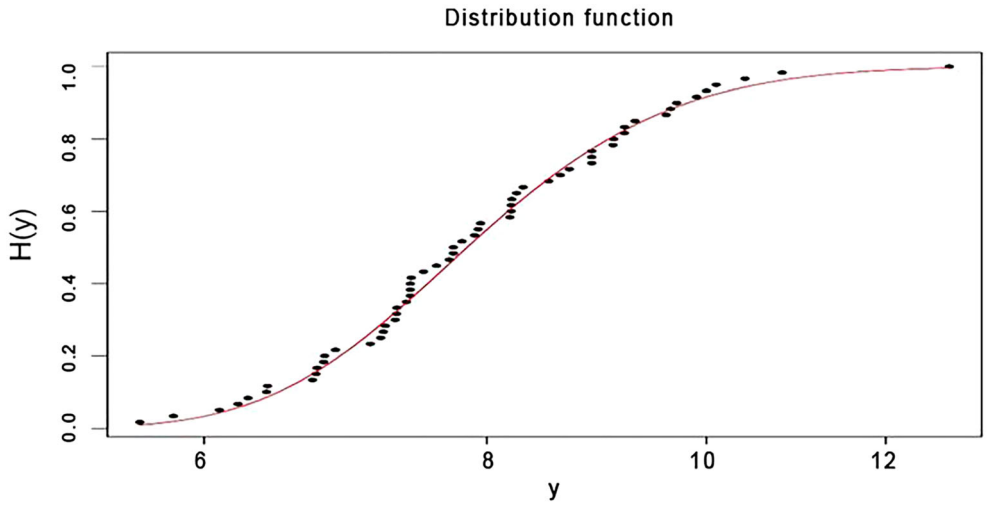


Figure 8. Fitting of distribution function.

respectively. However, the SVM model predicts the wave height with higher accuracy of 99.80%. The result suggests that the ML model performs better in the Mehamn harbour region. However, the SVM model gives the best result with higher accuracy. Based on the comparison results, it can be concluded that the SVM model outperforms the LR and ANN models in the present study.

Table 2 shows the comparison of the performance of the SVM model with varying kernels. The multiquadric and rational quadratic kernel function of the SVM model does not give satisfactory prediction results. However, inverse multiquadric, Laplacian, and Gaussian kernel functions give better prediction results. Out of various kernel functions, the Gaussian kernel function performs prediction exceptionally well with 99.80% accuracy. The comparison results showed that the Gaussian kernel function used in the SVM model outperforms the various existing kernel functions.

From tables 1 and 2, it can be observed that SVM (with Gaussian RBF) provides relatively better results among various ML models. Therefore, the SVM model was used for the prediction of significant wave height for the Mehamn harbour data. The SVM approach to this data resulted in the prediction of significant wave height with an RMSE value of 3.95 cm. Since the value of the significant wave height of the dataset lies between 0 and 12.6 m, considering a significant wave height of 6 m, RMSE of 3.95 cm corresponds to 0.7%. The dataset contains a large number of observations therefore overfitting of the ML model has to be taken care of. In order to have a more accurate model and avoid overfitting, K-fold cross-validation was performed. The current data was split into five folds and analysis yielded an average R^2 value of 0.998 (individual R^2 values: 0.998, 0.997, 0.996, 0.998 and 0.997).

A range of values of SVM model parameters (gamma and epsilon) was tested and the model was fine-tuned to yield the best results. Table 3 shows the range of values of SVM model parameters (gamma and epsilon) that were tested and tuned

Table 1. *Comparison of different machine learning models.*

Model	RMSE (cm)	R^2
Linear regression (LR)	10.8	0.989
Artificial neural network (ANN)	7.8	0.994
Support vector machine (SVM)	3.95	0.998

Table 2. *Comparison of different kernels for SVM model.*

Kernel	RMSE (cm)	R^2
Gaussian	3.95	0.9980
Laplacian	10.95	0.9886
Inverse multiquadric	39.91	0.8526
Rational quadratic	69.98	0.5483
Multiquadric	104.41	–0.0081

Table 3. *SVM parameters that yielded RMSE value of 3.95 cm.*

SVM parameters	Value
Gamma	0.011
Epsilon	0.01
Kernel	Gaussian radial basis function

fine to yield the best results. The optimum value of the gamma and epsilon parameters used in the SVM model is 0.011 and 0.01, respectively. The optimum value of the gamma and epsilon parameters in the SVM model is used to give a higher R^2 value of 0.998. The R^2 value can be justified by figure 9 which shows a plot between predicted values of significant wave height and actual significant wave height.

Once the prediction of significant wave height is completed, the later part of the work includes the analysis of extreme wave heights. The analysis of extreme wave height combines ideas from both ML as well as statistics, since statistics define the model and ML helps to calibrate such models. In statistics, the GEV theory deals with the behaviour of the extreme values of the dataset. Initially, the extreme maximum wave height of each year (i.e., 60) is extracted from the dataset using the block maxima (BM) technique, and the GEV distribution is fitted to the extracted yearly maximum wave height dataset. The best-fitting distribution of these sixty (60) maxima points yields the value of the shape (ε), scale (μ), and location parameters (σ). These values of parameters were estimated based on maximum likelihood estimation since the observations for significant wave height were known (Myung 2003). In the process, a likelihood function is defined and the iterations are performed in order to maximize the likelihood function and the values of parameters at maxima are the maximum likelihood estimators. These value of the shape parameter corresponds to the Weibull distribution. Based on these values of GEV parameters, equation (3) is used to calculate the return level (i.e., extreme wave height (Z_T)) with a return period of 'T' years. For example, Shajitha and Perera (2014) predicted 100 yr return period of extreme sea wave heights from a 30-yr dataset using GEV distribution in Colombo, Sri Lanka.

In this study, first, the training dataset (50% of the data, i.e., 1957–1986 data) and testing dataset (50% of the data, i.e., 1987–2017 data) had been used to estimate the return period of 100 yrs separately. The training and testing period of extreme wave height with respect to the return period event from the time series dataset are shown in figures 10 and 11.

A comparison of extreme wave height with a return period of 5, 20 and 100 yrs during the training and testing period is shown in table 4. The extreme wave height with a return period of 5, 20 and 100 yrs during the training and testing period of the model development are 2.389, 3.568 and 4.875 m, respectively. However, the extreme wave height in the testing phase with a return period of 5, 20 and 100 yrs are 2.464, 3.652 and 4.969, respectively.

It can be observed that the prediction pattern of extreme wave height using the training and testing data is almost similar. That shows the training dataset is in good agreement with the testing dataset.

Further, the predicted dataset from the machine learning model is used to analyze the extreme wave height of the return period of 100 yrs. The GEV distribution is fitted to the 60-yr dataset of predicted values of the machine learning model. From

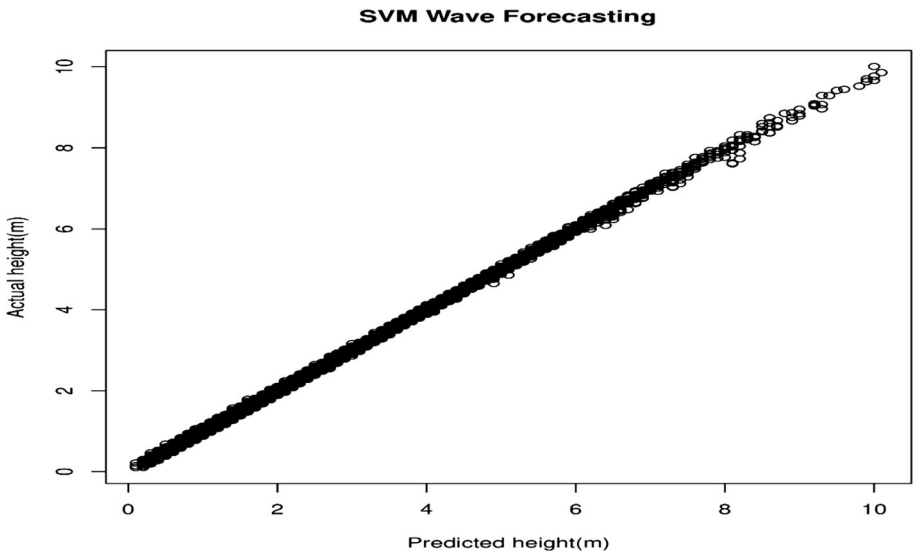


Figure 9. Comparison of actual *vs.* predicted significant wave height.

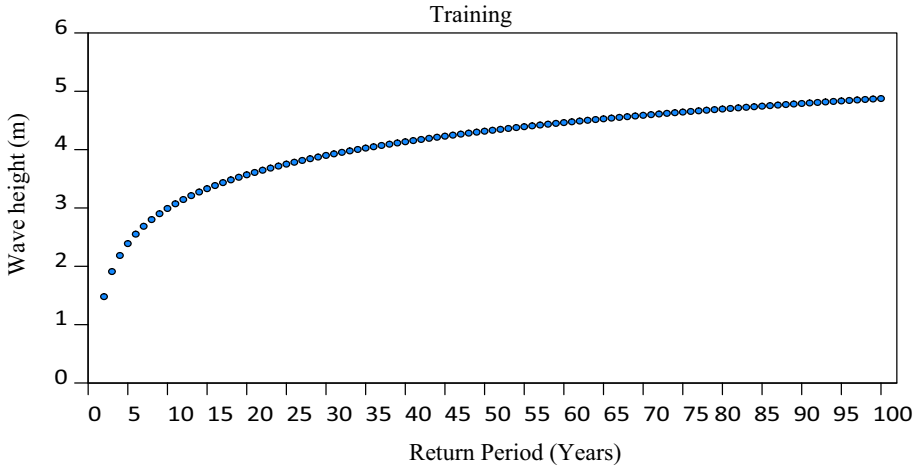


Figure 10. Extreme wave height with return period in training phase.

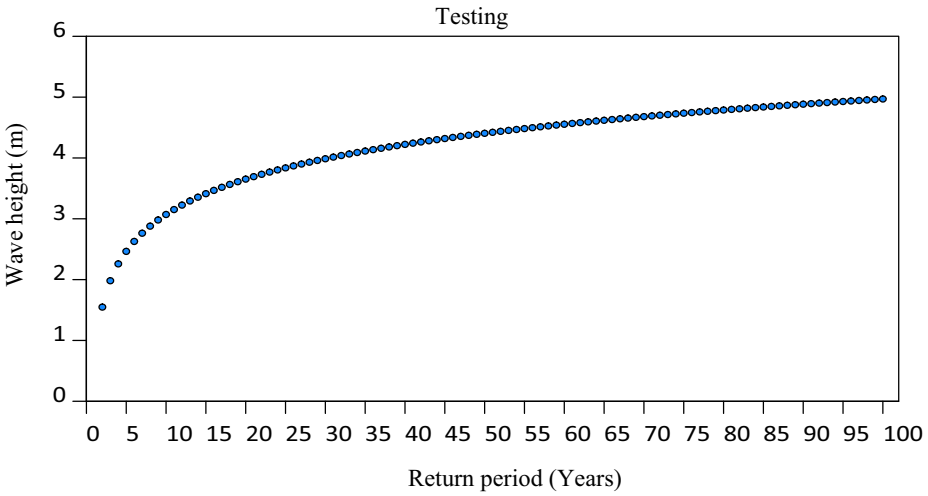


Figure 11. Extreme wave height with return period in testing phase.

Table 4. *Comparison of extreme wave height with return period of 5, 20 and 100 yrs during the training and testing period.*

Return period (yr)	Extreme wave height	
	Training	Testing
5	2.389	2.464
10	2.991	3.070
20	3.568	3.652
50	4.315	4.405
75	4.643	4.735
100	4.875	4.969

maximum likelihood estimation, the value of the shape, location and scale parameters is calculated, shown in table 5.

Table 5 shows the values resulting from maximum likelihood estimation. The scale parameter

Table 5. *Values resulted from maximum likelihood estimation.*

Parameters	Values
Shape parameter	–0.054
Location parameter	7.432
Scale parameter	1.124

and the location parameters give a positive value of 1.124 and 7.432, respectively. However, the shape parameter indicates a negative value of −0.054. The negative value of the shape parameter suggests that the extreme values follow a Weibull distribution. From the obtained values of GEV parameters, return levels of the extreme waves are predicted along with their return period. These values of return levels are plotted against the

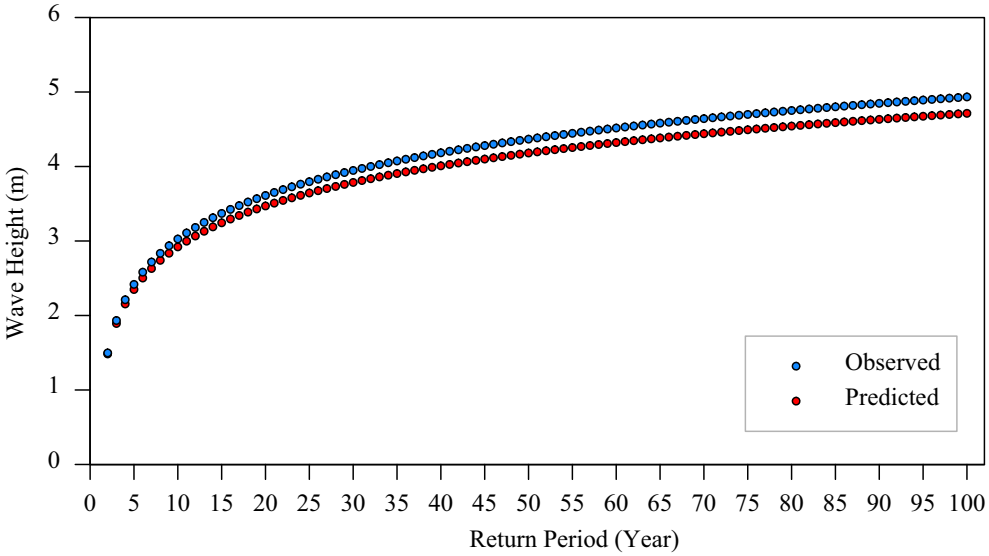


Figure 12. Comparison of extreme wave height along with its return period.

Table 6. *Comparison of predicted wave height with observed wave height.*

Return period (yr)	Observed wave height (m)	Predicted wave height (m)
5	2.417	2.348
10	3.026	2.921
20	3.610	3.470
50	4.367	4.180
75	4.698	4.492
100	4.933	4.713

return period and are shown in figure 12. The predicted wave height are compared with the observed wave height for the different return period. The predicted wave height shows good agreement with the observed value that shows the correctness of the used methodology.

Table 6 represents the extreme wave height with a return period of 5, 20 and 100 yrs are 2.348, 3.470 and 4.713 m, respectively. A coastal structure is designed such that it is capable of dealing with the most extreme waves during its design life, without being affected by waves. Quantitatively speaking, in our study for a structure with a design life of 100 yrs, it must be capable of dealing with waves of height 4.713 m.

4.1 *Significance of prediction using machine learning*

The machine learning (ML) model (which is mainly data-driven) in the present work has been used for significant wave height prediction. It is indeed true that a well-made physics-based model (WAM) enables us to understand complex processes and predict future events. Although the WAM models performed reasonably well in some of the cases, in rapidly varying circumstances, these models simply failed to give a proper description of the sea state. In addition, the WAM model did not provide better prediction results in the prediction of significant wave height (Janssen 2003). To overcome these limitations, the ML models were incorporated with WAM-generated data in the present study. This is an acceptable and verified approach as described by Khlongkhoi (2019) and Lou *et al.* (2021).

There is a growing consensus that solutions to complex science and engineering problems require novel methodologies that are able to integrate traditional physics-based modelling approaches with state-of-the-art ML techniques. Generally, ML models learn from experience in principle resembles the way humans learn. A class of ML models called artificial neural networks are computing systems inspired by how the brain processes information and learns from experience. The same concept of inherent learning of physics involved in wave prediction is done by training ML models in the current study. The input parameters that have been used for training the ML models are the real data comprising the most important wave parameters generated using the physics-based model WAM.

The physics-based model's simulated data have been consistently used to pre-train the ML model

(Tur *et al.* 2017; Jia *et al.* 2019). In one of the studies, Read *et al.* (2019) demonstrated that such models are able to generalize better to unseen scenarios than pure physics-based models. In the present study, the physics-based dataset of wind, wind sea and swell parameters are generated from the WAM model. Authors have simply used the generated data in ML models for future predictions and to show that ML can be successfully used for future predictions. This is a verified approach and is the same as Browne *et al.* (2007), where they used a similar input parameters model set (comprising of significant wave height, wind, wind sea and swell parameters) to their ANN model at a different location. Therefore, the present study attempts definitive testing of ML for modelling the significant wave height; bringing global ocean wave model output to nearshore locations, and demonstrating a potentially useful tool for emulating expensive wave forecasting observations.

5. Conclusion

The primary objective of the current study is to predict significant wave heights including extreme wave analysis using ML tools. The data used in the study belongs to Mehamn harbour, Norway that is obtained from ECMWF. The data is analyzed using various ML tools such as ANN, LR and SVM that predict significant wave height based on wind, wind sea and swell parameters. The presented model has been trained with the current data and fine-tuned (specific values of gamma, epsilon and chosen Kernel for SVM) for the location at hand, i.e., Mehamn. However, the SVM model can be fine-tuned for any other location worldwide as long as the data is available for that area. The LR and ANN models predict wave height with an accuracy of 98.90% and 99.40%, respectively. However, the SVM model predicts the significant wave height with higher accuracy of 99.80%. Based on the comparison results, it can be concluded that the SVM model outperforms the LR and ANN models in the present study.

Further, in order to estimate the extreme waves along with their return period, the probability distribution function (PDF) was identified using the GEV theory. The block maxima approach was used in order to tackle the variations due to seasonality. The shape parameter (ε), location parameter (μ) and scale parameter (σ) of GEV distribution are predicted by maximum likelihood estimation that signifies a Weibull distribution of the extreme values ($\varepsilon = -0.054$, $\mu = 7.432$, $\sigma = 1.124$). It is observed that the values of extreme wave height with a return period of 5, 20, and 100 years are 2.348, 3.470 and 4.713 m, respectively.

The results obtained from the current study will be highly useful for the optimal design of the ocean structures in Mehamn. Similar studies can be conducted at different other locations worldwide and the results hence obtained can be used for the optimal design of ocean structures in those areas. It is noteworthy to mention that the prediction time for obtaining these values from a physics-based model is very high.

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Author statement

MSA: Conceptualisation of work, methodology, manuscript preparation, supervision and revision of the draft manuscript, LK: Data collection, numerical modelling, data analysis and interpretation of results and drafting the original and revised manuscript. VC and YK: Model analysis, numerical modelling and data analysis. MZ: Data analysis and interpretation, and drafting the revised manuscript.

Appendix A

A.1 Machine learning models

A.1.1 Linear regression (LR)

The linear regression method generally formulates the output as a linear function of the input variables in such a way that the values obtained using the formulated function are close to the actual output. This formulated linear function (hypothesis) can be further used to predict the output values for any given input. The method of LR involves creating a cost function which is a function of these parameters (constants). The lesser the value of the cost function the lower the difference between the

predicted values and the actual values. Hence, the values of these parameters are determined by minimizing the value of the cost function. This is achieved using optimization techniques such as gradient descent.

A.1.2 Artificial neural network (ANN)

An artificial neural network (ANN) consists of an input layer, a series of hidden layers and an output layer, each having a various number of nodes. The input layer contains an equal number of nodes as input features. Similarly, the output layer is the same as the number of outputs desired by the model, which is generally one for regression models. The hidden layers contain a number of nodes that act as a parameter for the model.

In this neural network technique, the input layer uses feature value as input. The input layer contains an activation function (such as relu, tanh, or sigmoidal), which allow input to pass through it and provides the output from the input layer. The input layer is connected with the hidden layer with some weight factor. The weight factor determines the relative contribution of different nodes in the hidden layer. The output obtained from the input layer act as an input to the hidden layer. Further, the activation function determines the output of the hidden layer. Now, the hidden layer output is multiplied with the subsequent weight factor to obtain input to the output layer. Therefore, the final output is obtained by adding all the input of the output layer. After the first iteration, the weight is updated, and the processes are repeated until the minimum difference is obtained from the actual and predicted output, this process is also called back propagation algorithm.

The multilayer perceptron neural network is used for this study, which comprises three hidden layers with 10, 20 and 30 nodes, respectively. The number of nodes in each layer is decided by performing iterations for a different number of nodes in each layer. The relu (rectified linear unit) activation function was used for training the model. The other two widely used activation functions, namely tanh and logistic activation functions have output values in the range of $[-1, 1]$ and $[0, 1]$, respectively, and hence are more suitable for classification problems. As this is not the case with relu function, it is used as the activation function for training the model.

A.1.3 Support vector machine (SVM)

Support vector machine (SVM) algorithms employ a number of mathematical formulations, which are defined as the kernel. The kernel's function uses input data and converts them into the required form. Basically, it returns the internal product between two points in an appropriate feature space. The linear, non-linear, polynomial, and radial basis function is used as kernel function in the SVM algorithm (Gunn *et al.* 1998). The kernel function gained considerable attention in the last few years, especially as the SVM algorithm gained popularity. In many applications, kernel functions provide a simple relationship between linearity and non-linearity for the algorithms, which can be expressed as dot products (Fadel *et al.* 2016). In this paper, the Gaussian radial basis function kernel (popular kernel function used in various kernelized learning algorithms) has been used for the prediction of significant wave height using the SVM model as it outperformed other RBF kernels by significant difference (comparison is shown in the results section).

Gamma (γ) is an important parameter of the Gaussian RBF kernel. The gamma parameter depicts the extent to which an individual training example influences, with low values that mean 'far' and high values that mean 'close'. The gamma parameters can be viewed as the inverse of the sample's influence radius selected as support vectors by the model. Analytically, the narrower the Gaussian RBF kernels get (larger gammas) the 'spikier' the hypersurface is going to get. On the other hand, if the Gaussian RBF kernels are too wide (small gammas), it would end up with a hypersurface that is almost flat. Hence, choosing an optimum value of gamma is very important.

The advantages of the RBF are ease of construction, good generalization, high input noise tolerance, and the capacity to learn. The radial basis function can be used to develop the SVM model as well as ANN Model. When dealing with a large number of high-dimensional datasets, the ANN with RBF techniques is ineffective due to their sensitivity to the Hughes phenomenon. SVM with radial basis function has been effectively applied for high dimensional datasets in recent years (Huang *et al.* 2002; Camps-Valls *et al.* 2004;

Melgani and Bruzzone 2004). The properties of SVM with RBF kernel allow them to deal with a large set of high-dimensional datasets. The SVM with RBF kernel can handle large input spaces, cope with noisy samples robustly and create sparse solutions effectively.

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